AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Basics of time series and data structures



Session 1 of 10

Outline

- Introduction to forecasting
- Time series data
- tsibble objects
- Example: Australian prison population
- Example: Australian pharmaceutical sales

SECTION 1 OF 5

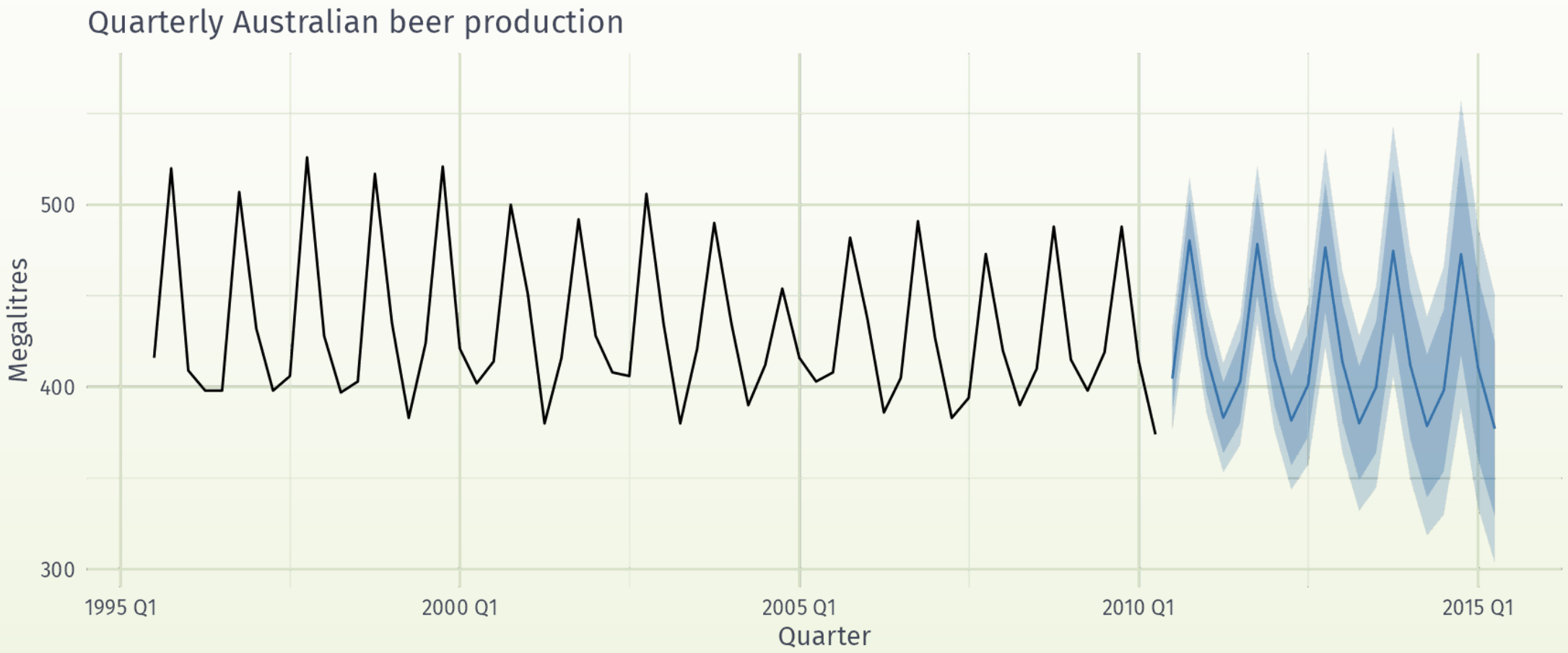
Introduction to forecasting

1 **Introduction to forecasting** 2 Time series data 3 tsibble objects 4 Example: Australian prison population 5 Example: Australian pharmaceutical sales



What is a forecast?

Estimating how a sequence of observations will continue into the future, using all information available now.



What data do we need?

1. Historical data on the variable to forecast
2. Past and future values of predictors
   - Deterministic (e.g. holidays)
   - Stochastic (e.g. temperature)
3. Expert knowledge and context
   - New information

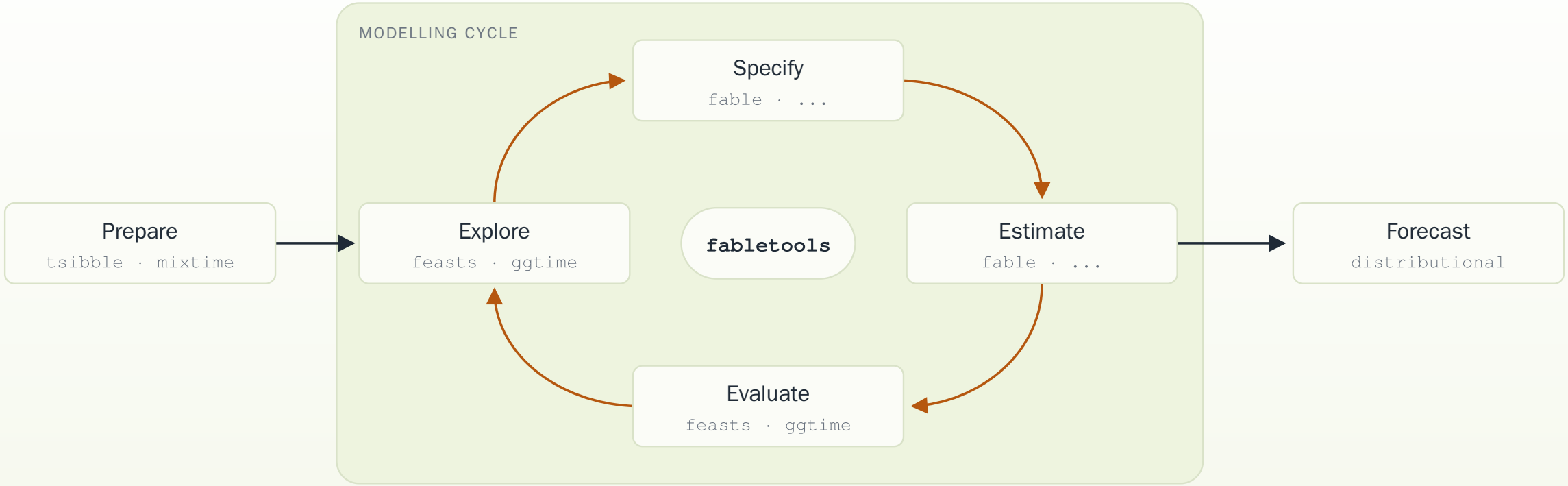
Why forecast?

Why do you use forecasts?

Decision	Forecast
Build a new hospital in the next 10 years?	Population and patient demand, years ahead
Call centre staff needed next week?	Call volumes, hours to days ahead
Vaccine doses needed next month?	Doses by region, weeks ahead

The decision sets **what** to forecast and **how far ahead**.

Forecasting workflow



Packages for this course

tidyverts.org



Time series data



Example datasets



Features and decompositions



Time series graphics



Forecasting models

Packages for this course

```
# Data, graphics and forecasting
library(fpp3)
```

```
-- Attaching packages ------------------------------- fpp3 1.0.3 --
v tibble      3.3.1          v tsibble     1.2.0.9000
v dplyr       1.2.1          v tsibbledata 0.4.1
v tidyr       1.3.2          v ggtime      1.0.0.9000
v lubridate   1.9.5          v feasts      0.5.0
v ggplot2     4.0.3          v fable       0.5.0.9000
```

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Time series data

1 Introduction to forecasting
2 **Time series data**
3 tsibble objects
4 Example: Australian prison population
5 Example: Australian pharmaceutical sales



Time series data

- Four-yearly Olympic winning times
- Annual Google profits
- Quarterly Australian beer production
- Monthly rainfall
- Weekly retail sales
- Daily IBM stock prices
- Hourly electricity demand
- 5-minute freeway traffic counts
- Time-stamped stock transaction data

Your time series

What time series data do you work with?

Waiting for answers...

poll.nectric.com.au/s/
africast-basics

0 responses

● CLOSED 0 connected

What makes it a time series?

Observations + **when** they happened.

- The time column is the **index**
- Observations are ordered in time
- Regular intervals define the frequency (yearly, quarterly, ...)

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tsibble objects

1 Introduction to forecasting
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A `tsibble` has three parts

A **tsibble** stores one or more time series in a table.

- **Index**: when
- **Key**: which series (optional)
- **Measures**: what was observed

Works with dplyr verbs (`filter()`, `mutate()`, ...).

`tsibble` objects

```
global_economy
```

```
# A tsibble: 15,150 x 6 [1Y]
# Key:       Country [263]
    Year Country          GDP Imports Exports Population
   <dbl> <fct>            <dbl>   <dbl>   <dbl>      <dbl>
 1  1960 Afghanistan  537777811.    7.02    4.13    8996351
 2  1961 Afghanistan  548888896.    8.10    4.45    9166764
 3  1962 Afghanistan  546666678.    9.35    4.88    9345868
 4  1963 Afghanistan  751111191.   16.9     9.17    9533954
 5  1964 Afghanistan  800000044.   18.1     8.89    9731361
 6  1965 Afghanistan 1006666638.   21.4    11.3     9938414
 7  1966 Afghanistan 1399999967.   18.6     8.57   10152331
 8  1967 Afghanistan 1673333418.   14.2     6.77   10372630
 9  1968 Afghanistan 1373333367.   15.2     8.90   10604346
10  1969 Afghanistan 1408888922.   15.0    10.1    10854428
# i 15,140 more rows
```

Index **Key** **Measures**

`tsibble` objects

```
tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:       Region, State, Purpose [304]
   Quarter Region   State Purpose  Trips
     <qtr> <chr>    <chr> <chr>    <dbl>
 1 1998 Q1 Adelaide SA    Business  135.
 2 1998 Q2 Adelaide SA    Business  110.
 3 1998 Q3 Adelaide SA    Business  166.
 4 1998 Q4 Adelaide SA    Business  127.
 5 1999 Q1 Adelaide SA    Business  137.
 6 1999 Q2 Adelaide SA    Business  200.
 7 1999 Q3 Adelaide SA    Business  169.
 8 1999 Q4 Adelaide SA    Business  134.
 9 2000 Q1 Adelaide SA    Business  154.
10 2000 Q2 Adelaide SA    Business  169.
# i 24,310 more rows
```

Index **Keys** **Measure**

Overnight trips (thousands), quarterly, by region and purpose of travel.

The `tsibble` index

For annual data, integer years work as the index:

```
annual_data <- tsibble(
  year = 2012:2016,
  y = c(123, 39, 78, 52, 110),
  index = year
)
annual_data
```

```
# A tsibble: 5 x 2 [1Y]
   year     y
  <int> <dbl>
1  2012   123
2  2013    39
3  2014    78
4  2015    52
5  2016   110
```

The `tsibble` index

```
monthly_data
```

```
# A tibble: 5 × 2
  Month    Observation
  <chr>          <dbl>
1 2019 Jan          50
2 2019 Feb          23
3 2019 Mar          34
4 2019 Apr          30
5 2019 May          25
```

- `"2019 Jan"` is just text
- More frequent than yearly? Use a time class

The `tsibble` index

```
monthly_data |>
  mutate(Month = yearmonth(Month)) |>
  as_tsibble(index = Month)
```

```
# A tsibble: 5 x 2 [1M]
     Month Observation
     <mth>       <dbl>
1 2019 Jan          50
2 2019 Feb          23
3 2019 Mar          34
4 2019 Apr          30
5 2019 May          25
```

The `tsibble` index

Common time index variables can be created with these functions:

Frequency	Function
Annual	`start:end`
Quarterly	`yearquarter()`
Monthly	`yearmonth()`
Weekly	`yearweek()`
Daily	`as_date()`, `ymd()`
Sub-daily	`as_datetime()`, `ymd_hms()`

Converting dates

Often data is provided as dates, even if it isn't daily.

```
my_date <- as.Date("2026-10-01")
yearquarter(my_date)
```

```
<yearquarter[1]>
[1] "2026 Q4"
# Year starts on: January
```

```
yearmonth(my_date)
```

```
<yearmonth[1]>
[1] "2026 Oct"
```

```
yearweek(my_date)
```

```
<yearweek[1]>
[1] "2026 W40"
# Week starts on: Monday
```

Looking ahead: `mixtime`



One family of linear time classes, including `year()` and `datetime()`:

```
library(mixtime)
c(year(my_date), yearquarter(my_date), yearmonth(my_date), date(my_date))
```

```
<mixtime[4]>
[1] 2026       2026 Q4    2026 Oct   2026-10-01
```

Any granularity: pick a **chronon** (one time step) and call `linear_time()`:

```
quarter_hours <- datetime("2024-01-01 00:00:00") + minutes(0:3) * 15L
linear_time(quarter_hours, chronon = cal_gregorian$minute(15L))
```

```
<mixtime[4]>
[1] 2024-01-01 00:00 2024-01-01 00:15 2024-01-01 00:30 2024-01-01 00:45
```

- `yearmonth(my_date)` is `linear_time(my_date, chronon = cal_gregorian$month(1L))`

Looking ahead: `mixtime`



More time types, beyond this course:

```
# Timezones
datetime(as.POSIXct("2026-10-05 22:30", tz = "UTC"), tz = "Africa/Nairobi")
```

```
<mixtime[1]>
[1] 2026-10-06 01:30:00 EAT
```

```
# Other calendars (also cal_isoweek, lunar and solar calendars)
linear_time(my_date, chronon = cal_sym454$month(1L))
```

```
<mixtime[1]>
[1] 2026-Oct
```

```
# Cyclical time
day_of_week(my_date)
```

```
<mixtime[1]>
[1] Thu
```

```
# Durations
yearmonth(my_date) + months(0:2)
```

```
<mixtime[3]>
[1] 2026 Oct 2026 Nov 2026 Dec
```

What a `tsibble` checks

- Index + key must **uniquely identify** each row: `as_tsibble()` errors on duplicates (find them with `duplicates()`)
- The **interval** is shown in the header: `[1Y]`, `[1Q]`, `[1M]`, ...
- Missing time steps are **implicit gaps**: find them with `has_gaps()`, make them explicit with `fill_gaps()`

```
tsibble(year = c(2012, 2013, 2015, 2016), y = c(123, 39, 52, 110), index = year) |>
  fill_gaps()
```

```
# A tsibble: 5 x 2 [1Y]
   year     y
  <dbl> <dbl>
1  2012   123
2  2013    39
3  2014    NA
4  2015    52
5  2016   110
```

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Example: Australian prison population

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Australian prison population



Read a csv file

Quarterly prisoner counts from the ABS, 2005 Q1 to 2016 Q4, by state, gender, legal status and indigenous status (ATSI: Aboriginal and Torres Strait Islander).

```
prison <- readr::read_csv("data/prison_population.csv")
```

```
# A tibble: 3,072 × 6
   date       state gender legal     indigenous count
   <date>     <chr> <chr>  <chr>     <chr>      <dbl>
 1 2005-03-01 ACT   Female Remanded  ATSI           0
 2 2005-03-01 ACT   Female Remanded  Other          2
 3 2005-03-01 ACT   Female Sentenced ATSI           0
 4 2005-03-01 ACT   Female Sentenced Other          0
 5 2005-03-01 ACT   Male   Remanded  ATSI           7
 6 2005-03-01 ACT   Male   Remanded  Other         58
 7 2005-03-01 ACT   Male   Sentenced ATSI           0
 8 2005-03-01 ACT   Male   Sentenced Other          0
 9 2005-03-01 NSW   Female Remanded  ATSI          51
10 2005-03-01 NSW   Female Remanded  Other        131
# i 3,062 more rows
```

Convert to a time class

```
prison <- readr::read_csv("data/prison_population.csv") |>
  mutate(quarter = yearquarter(date))
```

```
# A tibble: 3,072 × 7
   date       state gender legal     indigenous count quarter
   <date>     <chr> <chr>  <chr>     <chr>       <dbl>   <qtr>
 1 2005-03-01 ACT   Female Remanded  ATSI            0 2005 Q1
 2 2005-03-01 ACT   Female Remanded  Other           2 2005 Q1
 3 2005-03-01 ACT   Female Sentenced ATSI            0 2005 Q1
 4 2005-03-01 ACT   Female Sentenced Other           0 2005 Q1
 5 2005-03-01 ACT   Male   Remanded  ATSI            7 2005 Q1
 6 2005-03-01 ACT   Male   Remanded  Other          58 2005 Q1
 7 2005-03-01 ACT   Male   Sentenced ATSI            0 2005 Q1
 8 2005-03-01 ACT   Male   Sentenced Other           0 2005 Q1
 9 2005-03-01 NSW   Female Remanded  ATSI           51 2005 Q1
10 2005-03-01 NSW   Female Remanded  Other         131 2005 Q1
# i 3,062 more rows
```

Remove the old date column

```
prison <- readr::read_csv("data/prison_population.csv") |>
  mutate(quarter = yearquarter(date)) |>
  select(-date)
```

```
# A tibble: 3,072 × 6
   state gender legal     indigenous count quarter
   <chr> <chr>  <chr>     <chr>      <dbl>   <qtr>
 1 ACT   Female Remanded  ATSI           0 2005 Q1
 2 ACT   Female Remanded  Other          2 2005 Q1
 3 ACT   Female Sentenced ATSI           0 2005 Q1
 4 ACT   Female Sentenced Other          0 2005 Q1
 5 ACT   Male   Remanded  ATSI           7 2005 Q1
 6 ACT   Male   Remanded  Other         58 2005 Q1
 7 ACT   Male   Sentenced ATSI           0 2005 Q1
 8 ACT   Male   Sentenced Other          0 2005 Q1
 9 NSW   Female Remanded  ATSI          51 2005 Q1
10 NSW   Female Remanded  Other        131 2005 Q1
# i 3,062 more rows
```

Declare the index and keys

```
prison <- readr::read_csv("data/prison_population.csv") |>
  mutate(quarter = yearquarter(date)) |>
  select(-date) |>
  as_tsibble(
    index = quarter,
    key = c(state, gender, legal, indigenous)
  )
```

```
# A tsibble: 3,072 x 6 [1Q]
# Key:       state, gender, legal, indigenous [64]
   state gender legal    indigenous count quarter
   <chr> <chr>  <chr>    <chr>      <dbl>   <qtr>
 1 ACT   Female Remanded ATSI           0 2005 Q1
 2 ACT   Female Remanded ATSI           1 2005 Q2
 3 ACT   Female Remanded ATSI           0 2005 Q3
 4 ACT   Female Remanded ATSI           0 2005 Q4
 5 ACT   Female Remanded ATSI           1 2006 Q1
 6 ACT   Female Remanded ATSI           1 2006 Q2
 7 ACT   Female Remanded ATSI           1 2006 Q3
 8 ACT   Female Remanded ATSI           0 2006 Q4
 9 ACT   Female Remanded ATSI           0 2007 Q1
10 ACT   Female Remanded ATSI           1 2007 Q2
# i 3,062 more rows
```

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Example: Australian pharmaceutical sales

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Australian Pharmaceutical Benefits Scheme



Australian Pharmaceutical Benefits Scheme

The **PBS** is the Australian government drug subsidy scheme.

- Subsidises many pharmacy drugs
- Cost the government A$19.1 billion in 2024–25 (about 0.7% of GDP)
- Budget relies on forecasts of drug usage

`PBS` data: monthly, July 1991 to June 2008

- `Scripts` (prescriptions) and `Cost` (A$)
- By drug type (ATC2 x84), concession (x2) and patient type (x2)
- ATC: Anatomical Therapeutic Chemical classification

Working with `tsibble` objects

```
PBS
```

```
# A tsibble: 67,596 x 9 [1M]
# Key:       Concession, Type, ATC1, ATC2 [336]
      Month Concession   Type        ATC1  ATC1_desc                    ATC2  ATC2_desc Scripts  Cost
      <mth> <chr>        <chr>       <chr> <chr>                        <chr> <chr>       <dbl> <dbl>
 1 1991 Jul Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   18228 67877
 2 1991 Aug Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   15327 57011
 3 1991 Sep Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   14775 55020
 4 1991 Oct Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   15380 57222
 5 1991 Nov Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   14371 52120
 6 1991 Dec Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   15028 54299
 7 1992 Jan Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   11040 39753
 8 1992 Feb Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   15165 54405
 9 1992 Mar Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   16898 61108
10 1992 Apr Concessional Co-payments A     Alimentary tract and metab… A01   STOMATOL…   18141 65356
# i 67,586 more rows
```

`filter()`: choose rows

```
PBS |>
  filter(ATC2 == "A10")
```

```
# A tsibble: 816 x 9 [1M]
# Key:       Concession, Type, ATC1, ATC2 [4]
      Month Concession   Type        ATC1  ATC1_desc                    ATC2  ATC2_desc Scripts   Cost
      <mth> <chr>        <chr>       <chr> <chr>                        <chr> <chr>       <dbl>  <dbl>
 1 1991 Jul Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   89733 2.09e6
 2 1991 Aug Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   77101 1.80e6
 3 1991 Sep Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   76255 1.78e6
 4 1991 Oct Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   78681 1.85e6
 5 1991 Nov Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   70554 1.69e6
 6 1991 Dec Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   75814 1.84e6
 7 1992 Jan Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   64186 1.56e6
 8 1992 Feb Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   75899 1.73e6
 9 1992 Mar Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   89445 2.05e6
10 1992 Apr Concessional Co-payments A     Alimentary tract and meta… A10   ANTIDIAB…   97315 2.23e6
# i 806 more rows
```

`select()`: choose columns

```
PBS |>
  filter(ATC2 == "A10") |>
  select(Month, Concession, Type, Cost)
```

```
# A tsibble: 816 x 4 [1M]
# Key:       Concession, Type [4]
      Month Concession   Type           Cost
      <mth> <chr>        <chr>         <dbl>
 1 1991 Jul Concessional Co-payments 2092878
 2 1991 Aug Concessional Co-payments 1795733
 3 1991 Sep Concessional Co-payments 1777231
 4 1991 Oct Concessional Co-payments 1848507
 5 1991 Nov Concessional Co-payments 1686458
 6 1991 Dec Concessional Co-payments 1843079
 7 1992 Jan Concessional Co-payments 1564702
 8 1992 Feb Concessional Co-payments 1732508
 9 1992 Mar Concessional Co-payments 2046102
10 1992 Apr Concessional Co-payments 2225977
# i 806 more rows
```

`summarise()`: combine over keys

```
PBS |>
  filter(ATC2 == "A10") |>
  select(Month, Concession, Type, Cost) |>
  summarise(total_cost = sum(Cost))
```

```
# A tsibble: 204 x 2 [1M]
      Month total_cost
      <mth>      <dbl>
 1 1991 Jul    3526591
 2 1991 Aug    3180891
 3 1991 Sep    3252221
 4 1991 Oct    3611003
 5 1991 Nov    3565869
 6 1991 Dec    4306371
 7 1992 Jan    5088335
 8 1992 Feb    2814520
 9 1992 Mar    2985811
10 1992 Apr    3204780
# i 194 more rows
```

`mutate()`: create variables

```
PBS |>
  filter(ATC2 == "A10") |>
  select(Month, Concession, Type, Cost) |>
  summarise(total_cost = sum(Cost)) |>
  mutate(cost_m = total_cost / 1e6)
```

```
# A tsibble: 204 x 3 [1M]
      Month total_cost cost_m
      <mth>      <dbl>  <dbl>
 1 1991 Jul    3526591   3.53
 2 1991 Aug    3180891   3.18
 3 1991 Sep    3252221   3.25
 4 1991 Oct    3611003   3.61
 5 1991 Nov    3565869   3.57
 6 1991 Dec    4306371   4.31
 7 1992 Jan    5088335   5.09
 8 1992 Feb    2814520   2.81
 9 1992 Mar    2985811   2.99
10 1992 Apr    3204780   3.20
# i 194 more rows
```

`cost_m` is the total cost in A$ millions.

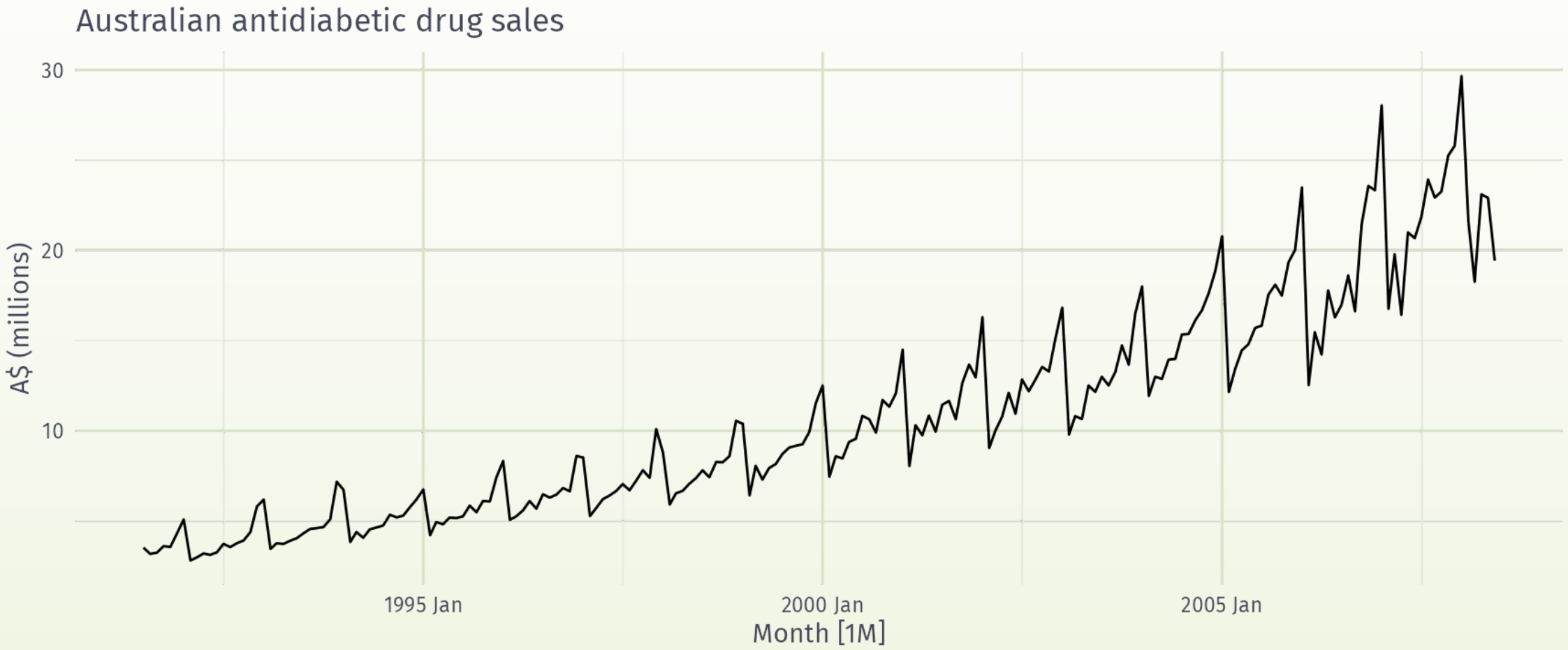
Save the result

```
a10 <- PBS |>
  filter(ATC2 == "A10") |>
  select(Month, Concession, Type, Cost) |>
  summarise(total_cost = sum(Cost)) |>
  mutate(cost_m = total_cost / 1e6)
```

```
# A tsibble: 204 x 3 [1M]
      Month total_cost cost_m
      <mth>      <dbl>  <dbl>
 1 1991 Jul    3526591   3.53
 2 1991 Aug    3180891   3.18
 3 1991 Sep    3252221   3.25
 4 1991 Oct    3611003   3.61
 5 1991 Nov    3565869   3.57
 6 1991 Dec    4306371   4.31
 7 1992 Jan    5088335   5.09
 8 1992 Feb    2814520   2.81
 9 1992 Mar    2985811   2.99
10 1992 Apr    3204780   3.20
# i 194 more rows
```

Plot the result

```
autoplot(a10, cost_m) +
  labs(y = "A$ (millions)", title = "Australian antidiabetic drug sales")
```



Your turn

Using `tourism`

1. Find the quarterly total `Trips` for each `State`
2. **Challenge:** Which `State` had the most `Trips` over the whole period?

Using `PBS`

3. Find the total monthly `Scripts` for ATC2 `"H02"`

Hints

- `summarise()` always keeps the index
- Q1: `group_by(State)`, then `summarise()`
- Q2: `as_tibble()` first: it drops the tsibble structure, so `summarise()` no longer keeps the index
- Q3: `filter()`, then `summarise()`

Recap

- A tsibble has an **index**, **keys** and **measures**
- Index + keys uniquely identify each row
- The index needs a time class (`yearmonth()`, `as_date()`, ...)
- Use dplyr verbs to wrangle tsibbles
- `summarise()` combines over keys, keeping the index